

①

Be: find the Laplace of $x(t) = e^{j\omega_0 t} u(t)$

$$\mathcal{L}\{x(t) = e^{j\omega_0 t} u(t)\} = \int_{-\infty}^{\infty} e^{j\omega_0 t} u(t) e^{-st} dt$$

$$= \int_0^{\infty} e^{-(s-j\omega_0)t} dt$$

$$= \left(\frac{1}{s-j\omega_0} \right)$$

Rez: - $\text{Re}(s) > 0$
 $\sigma > 0$

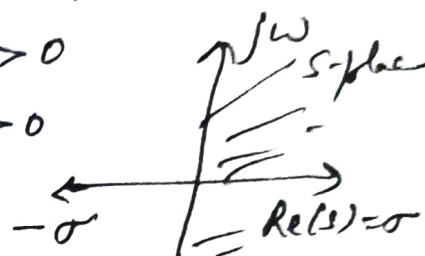
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Ques find the Laplace transform of following $x(t)$

(i) $x(t) = u(t)$

Solⁿ $X(s) = \mathcal{L}[x(t)] = \int_{-\infty}^{\infty} x(t) e^{-st} dt = \int_0^{\infty} 1 \cdot e^{-st} dt$

$= \frac{1}{s}$, RoZ:- $\text{Re}(s) > 0$
 $\Rightarrow \sigma > 0$



(ii) $x(t) = t u(t) \longrightarrow$ Ramp function

Solⁿ:- $X(s) = \mathcal{L}(t u(t)) = \int_{-\infty}^{\infty} t u(t) e^{-st} dt$

$= \int_0^{\infty} t e^{-st} dt = \left[t \frac{e^{-st}}{-s} \right]_0^{\infty} - \int_0^{\infty} \frac{e^{-st}}{-s} dt$

$= 0 + \frac{1}{s} \int_0^{\infty} e^{-st} dt = \frac{1}{s^2}$

RoZ:- $\text{Re}(s) > 0 \Rightarrow \sigma > 0$

$X(s) = \frac{1}{s^2}$, RoZ:- $\text{Re}(s) > 0$

(iii) $x(t) = \cos \omega t u(t)$

Solⁿ:- $x(t) = \cos \omega t u(t) = \frac{e^{j\omega t} + e^{-j\omega t}}{2} u(t)$

$\mathcal{L}[x(t)] = \int_{-\infty}^{\infty} x(t) e^{-st} dt = \int_0^{\infty} \frac{e^{j\omega t} + e^{-j\omega t}}{2} e^{-st} dt$

$$\begin{aligned}
 \mathcal{L}[x(t)] &= \frac{1}{2} \left[\int_0^{\infty} e^{-(s-j\omega)t} dt + \int_0^{\infty} e^{-(s+j\omega)t} dt \right] \quad (2) \\
 &= \frac{1}{2} \left[\frac{1}{s-j\omega} + \frac{1}{s+j\omega} \right] \quad \text{Roc: } -\text{Re}(s) > 0 \\
 &= \frac{s}{s^2 + \omega^2}
 \end{aligned}$$

So $\boxed{\cos \omega t \, u(t) \xleftrightarrow{\mathcal{L}} \frac{s}{s^2 + \omega^2}}$, Roc: $\text{Re}(s) > 0$

(iv) $x(t) = \sin \omega t \, u(t)$

Soln:- $x(t) = \sin \omega t \, u(t) = \left(\frac{e^{j\omega t} - e^{-j\omega t}}{2j} \right) u(t)$

$$= \frac{1}{2j} \left[e^{j\omega t} u(t) - e^{-j\omega t} u(t) \right]$$

$$\mathcal{L}[\sin \omega t \, u(t)] = \int_{-\infty}^{\infty} \sin \omega t \, u(t) e^{-st} dt$$

$$= \frac{1}{2j} \left[\int_0^{\infty} e^{j\omega t} e^{-st} dt - \int_0^{\infty} e^{-j\omega t} e^{-st} dt \right]$$

$$= \frac{1}{2j} \left[\int_0^{\infty} e^{-(s-j\omega)t} dt - \int_0^{\infty} e^{-(s+j\omega)t} dt \right]$$

$$= \frac{1}{2j} \left[\frac{1}{s-j\omega} - \frac{1}{s+j\omega} \right] \quad \text{Roc: } -\text{Re}(s) > 0$$

$$= \frac{1}{2j} \left[\frac{s+j\omega - s+j\omega}{s^2 + \omega^2} \right] = \frac{\omega}{s^2 + \omega^2}$$

(3) (4)

So

$$\sin \omega t u(t) \xleftrightarrow{\quad} \frac{\omega}{s^2 + \omega^2}, \quad \underline{\text{Re}(s) > 0}$$

(v) Hyperbolic sine and cosine functions
 $x(t) = \sinh \omega t u(t)$

Solⁿ:- $\sinh \omega t = \frac{e^{\omega t} - e^{-\omega t}}{2}$

$$\mathcal{L}[x(t)] = \mathcal{L}[\sinh \omega t u(t)] = \oint \int_{-\infty}^{\infty} \sinh \omega t u(t) e^{-st} dt$$

$$= \int_{-\infty}^{\infty} \frac{e^{\omega t} - e^{-\omega t}}{2} u(t) e^{-st} dt$$

$$= \frac{1}{2} \left[\int_0^{\infty} e^{-(s-\omega)t} dt - \int_0^{\infty} e^{-(s+\omega)t} dt \right]$$

$$= \frac{1}{2} \left[\frac{1}{s-\omega} - \frac{1}{s+\omega} \right] = \frac{\omega}{s^2 - \omega^2}, \quad \boxed{\text{for } \text{Re}(s) > 0}$$

So $\sinh \omega t \xleftrightarrow{\quad} \frac{\omega}{s^2 - \omega^2}$

(vi) $x(t) = \cosh \omega t u(t)$

$$\cosh \omega t = \frac{e^{\omega t} + e^{-\omega t}}{2}$$

$$\cosh \omega t u(t) \xleftrightarrow{\quad} \frac{s}{s^2 - \omega^2}, \quad \underline{\text{Re}(s) > 0}$$

(5)

Note:-

$$(i) \quad x(t) = -t u(t) \xrightarrow{\mathcal{L}} \frac{1}{s^2}, \quad \text{Re}(s) < 0$$

$$(ii) \quad x(t) = -\cos \omega t u(t) \xrightarrow{\mathcal{L}} \frac{s}{s^2 + \omega^2}, \quad \text{Re}(s) < 0$$

$$(iii) \quad x(t) = -\sin \omega t u(t) \xrightarrow{\mathcal{L}} \frac{\omega}{s^2 + \omega^2}, \quad \text{Re}(s) < 0$$

$$(iv) \quad x(t) = -\sinh \omega t u(t) \xrightarrow{\mathcal{L}} \frac{\omega}{s^2 - \omega^2}, \quad \text{Re}(s) < 0$$

$$(v) \quad x(t) = -\cosh \omega t u(t) \xrightarrow{\mathcal{L}} \frac{s}{s^2 - \omega^2}, \quad \text{Re}(s) < 0$$

Properties of Laplace transform:-

① Linearity Prop:-

$$y \quad x_1(t) \xrightarrow{\mathcal{L}} X_1(s), \quad \text{ROC: } \{R_1\}$$

$$x_2(t) \xrightarrow{\mathcal{L}} X_2(s), \quad \text{ROC: } \{R_2\}$$

$$\text{Then } a x_1(t) + b x_2(t) \xrightarrow{\mathcal{L}} a X_1(s) + b X_2(s)$$

$$\text{ROC: } R \in (R_1 \cap R_2)$$

② Time shifting Prop:-

$$y \quad x(t) \xrightarrow{\mathcal{L}} X(s), \quad \text{ROC: } \{R\}$$

$$\text{Then } x(t - t_0) \xrightarrow{\mathcal{L}} e^{-st_0} X(s), \quad \text{ROC: same as } X(s)$$

(6)

(3) shifting in s-domain

$$y \quad x(t) \xleftrightarrow{\mathcal{L}} X(s) \quad , \quad \text{ROC: } - \{R\}$$

$$\text{then } e^{-s_0 t} x(t) \xleftrightarrow{\mathcal{L}} X(s-s_0) \quad , \quad \text{ROC: } - \text{ROC}$$

Will be shifted
version of R

(4) Time scaling Prop:-

$$y \quad x(t) \xleftrightarrow{\mathcal{L}} X(s) \quad , \quad \text{ROC: } \{R\}$$

$$\text{then } x(at) \xleftrightarrow{\mathcal{L}} \frac{1}{|a|} X\left(\frac{s}{a}\right) \quad , \quad \text{ROC: } - \left\{\frac{R}{a}\right\}$$

scaled version of
old ROC.

(5) Complex conjugate Prop:-

$$y \quad x(t) \xleftrightarrow{\mathcal{L}} X(s) \quad , \quad \text{ROC: } \{R\}$$

$$\text{then } x^*(t) \xleftrightarrow{\mathcal{L}} X^*(s^*) \quad , \quad \text{ROC: Same as the ROC of } X(s)$$

i.e. $\{R\}$

(6) differentiation in time domain

$$y \quad x(t) \xleftrightarrow{\mathcal{L}} X(s) \quad , \quad \text{ROC: } \{R\}$$

$$\text{then } \frac{dx(t)}{dt} \xleftrightarrow{\mathcal{L}} sX(s) \quad , \quad \text{ROC: } - \text{at least } R$$

(7) Differentiation in s-domain

$$y \quad x(t) \xleftrightarrow{\mathcal{L}} X(s) \quad , \quad \text{ROC: } - \{R\}$$

$$\text{then } -tx(t) \xleftrightarrow{\mathcal{L}} \frac{d}{ds} X(s) \quad , \quad \text{ROC: } - \text{Same as } X(s)$$

(8)

Convolution Prop. in the domain

$$y \quad x_1(t) \xleftrightarrow{\mathcal{L}} X_1(s) \quad \& \quad x_2(t) \xleftrightarrow{\mathcal{L}} X_2(s)$$

$$\text{then } x_1(t) * x_2(t) \xleftrightarrow{\mathcal{L}} X_1(s) X_2(s)$$

(9)

Integration Prop:-

$$y \quad x(t) \xleftrightarrow{\mathcal{L}} X(s), \quad \text{ROZ: } \{R\}$$

$$\text{then } \int_{-\infty}^t x(\tau) d\tau \xleftrightarrow{\mathcal{L}} \frac{1}{s} X(s), \quad \text{ROZ:}$$

↓
ROZ will be at least
intersect the R and real
part of $s > 0$

(10)

Time Reversal Prop:-

$$y \quad x(t) \xleftrightarrow{\mathcal{L}} X(s)$$

$$\text{then } x(-t) \xleftrightarrow{\mathcal{L}} X(-s)$$

(11)

Parseval's theorem:

$$y \quad x_1(t) \xleftrightarrow{\mathcal{L}} X_1(s) \quad \& \quad x_2(t) \xleftrightarrow{\mathcal{L}} X_2(s)$$

$$\text{then } \int_{-\infty}^{\infty} x_1(t) x_2^*(t) dt = \frac{1}{2\pi j} \int_{-\infty}^{\infty} X_1(s) X_2^*(s) ds$$

Qe. find the Laplace transform of the following

(i)

$$x(t) = t e^{-t} u(t)$$

→ Hint (use differentiation Prop. in Laplace)

(ii)

$$x(t) = t e^{-(t-2)} u(t-2)$$

→ Hint (use the shifting Prop.)

(iii)

$$x(t) = e^{-3t} u(3t)$$

→ Hint (use the scaling Prop.)

(iv)

$$x(t) = t \cos \omega t u(t)$$

→ Hint (use differentiation Prop. in Laplace domain)

(v)

$$x(t) = t \sin \omega t u(t)$$

~~(vi) $x(t) = e^{at}$~~

Inverse Laplace transform using partial fraction: — ⑦

① ~~$\frac{1}{s}$~~ $\xleftrightarrow{\mathcal{L}^{-1}} u(t)$, $\text{Re}(s) > 0 \Rightarrow \sigma > 0$
 ~~$\frac{1}{s}$~~ $\xleftrightarrow{\mathcal{L}^{-1}} u(-t)$, $\text{Re}(s) < 0$

② $\frac{1}{s^2} \xleftrightarrow{\mathcal{L}^{-1}} t u(t)$, $\text{Re}(s) > 0$
 $\xleftrightarrow{\mathcal{L}^{-1}} -t u(-t)$, $\text{Re}(s) < 0$

③ $\frac{1}{(s+a)} \xleftrightarrow{\mathcal{L}^{-1}} e^{-at} u(t)$, $\text{Re}(s) > -a$
 $\xleftrightarrow{\mathcal{L}^{-1}} -e^{-at} u(-t)$, $\text{Re}(s) < -a$

④ $\frac{1}{(s-a)} \xleftrightarrow{\mathcal{L}^{-1}} e^{at} u(t)$, $\text{Re}(s) > a$
 $\xleftrightarrow{\mathcal{L}^{-1}} -e^{at} u(-t)$, $\text{Re}(s) < a$

⑤ $\frac{s}{s^2 + \omega^2} \xleftrightarrow{\mathcal{L}^{-1}} \cos \omega t u(t)$, $\text{Re}(s) > 0$
 $\xleftrightarrow{\mathcal{L}^{-1}} -\cos \omega t u(-t)$, $\text{Re}(s) < 0$

⑥ $\frac{\omega}{s^2 + \omega^2} \xleftrightarrow{\mathcal{L}^{-1}} \sin \omega t u(t)$, $\text{Re}(s) > 0$
 $\xleftrightarrow{\mathcal{L}^{-1}} -\sin \omega t u(-t)$, $\text{Re}(s) < 0$

⑦ $1 \xleftrightarrow{\mathcal{L}^{-1}} \delta(t)$

(10)

Qe1

Find the Inverse Laplace transform of

$$X(s) = \frac{1}{(s+4)(s-2)}$$

The region of convergence is

- (a) $-4 < \operatorname{Re}(s) < 2$ (b) $\operatorname{Re}(s) > 2$
 (c) $\operatorname{Re}(s) < -4$, (d) $2 < \operatorname{Re}(s) < -4$

Soln:-

$$\begin{aligned} X(s) &= \frac{1}{(s+4)(s-2)} = \frac{A}{s+4} + \frac{B}{s-2} \\ &= \frac{A(s-2) + B(s+4)}{(s+4)(s-2)} \end{aligned}$$

$$\text{So } A(s-2) + B(s+4) \equiv 1$$

→ To get value of A, put $s = -4$

$$A(-4-2) = 1 \Rightarrow A = -\frac{1}{6}$$

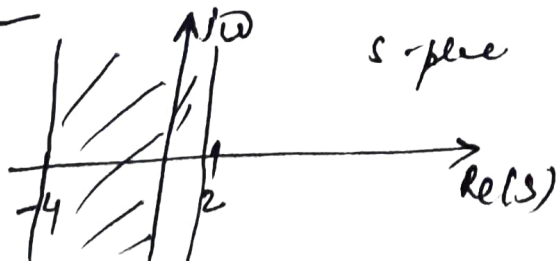
→ To get value of B, put $s = 2$

$$B(2+4) = 1 \Rightarrow B = \frac{1}{6}$$

$$\text{So } X(s) = \frac{1}{(s+4)(s-2)} = \frac{1}{6} \left[-\frac{1}{s+4} + \frac{1}{s-2} \right]$$

(a)

Ans: $-4 < \operatorname{Re}(s) < 2$
 s-plane

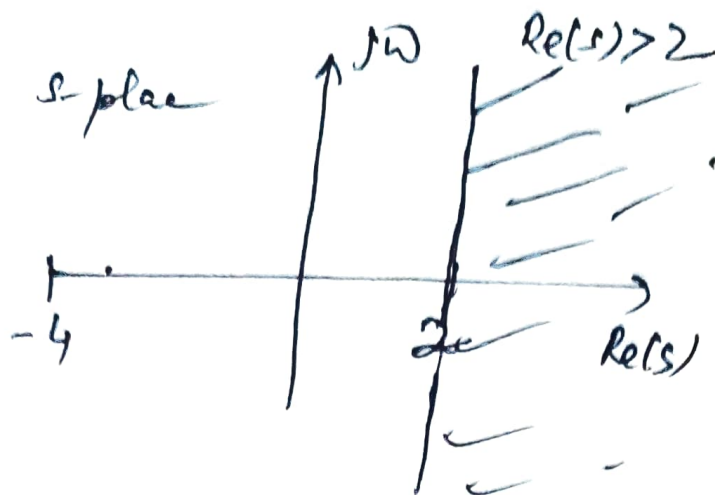


$$x(t) = \frac{1}{6} \left[-e^{-4t} u(t) - e^{2t} u(-t) \right]$$

(b)

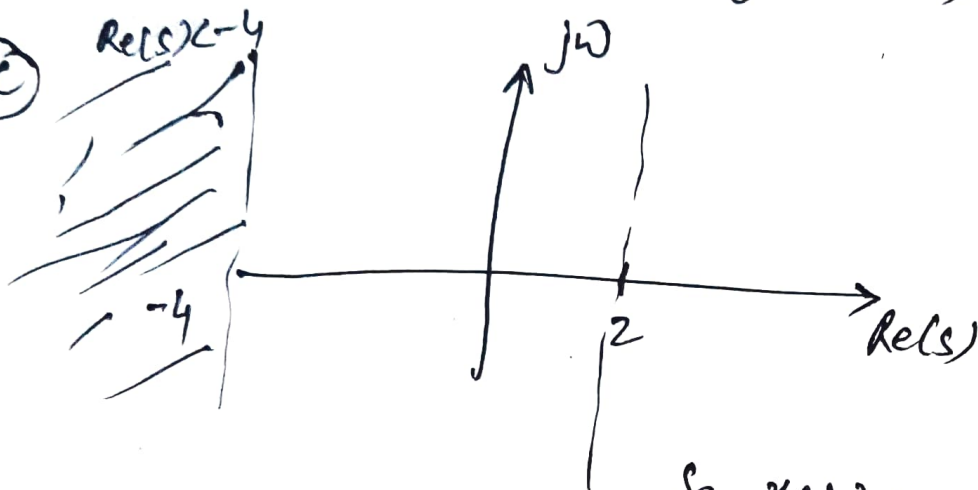
$$\operatorname{Re}(s) > 2$$

(11)



$$x(t) = -\frac{1}{6} e^{-4t} u(t) + \frac{1}{6} e^{2t} u(t)$$

(c)



$$\underline{\text{Roz:}} - \operatorname{Re}(s) < -4$$

$$\text{So } x(t) = -\frac{1}{6} e^{2t} u(-t) + \frac{1}{6} e^{-4t} u(-t)$$

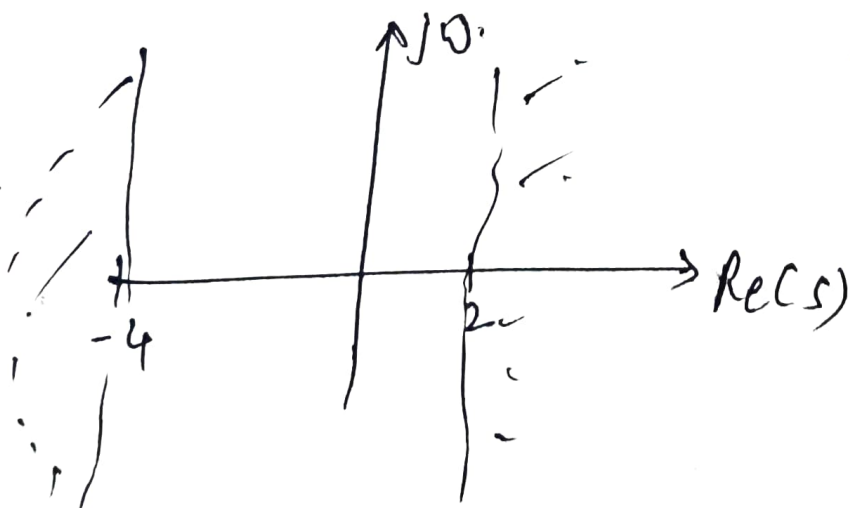
(d)

Roz:

$$2 < \operatorname{Re}(s) < -4$$

Ans: - This Roz is not possible

so $x(t)$ can't be determined

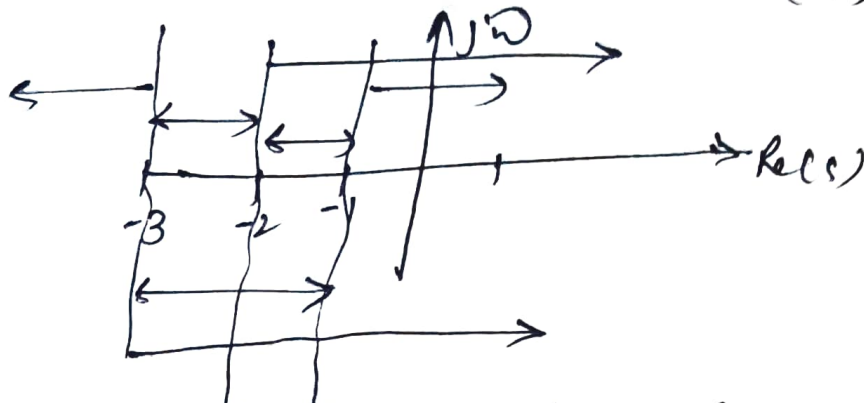


(12)

Ques Find all possible Inverse Laplace transform for $X(s) = \frac{1}{(s+1)(s+2)(s+3)}$

Solⁿ

$$X(s) = \frac{1}{(s+1)(s+2)(s+3)} = \frac{\frac{1}{2}}{(s+1)} - \frac{1}{(s+2)} + \frac{\frac{1}{2}}{(s+3)}$$



Possible case of ROC: (i) $\text{Re}(s) < -3$ ✓

(ii) $\text{Re}(s) > -3$ ✗

(vi) $-3 < \text{Re}(s) < -2$ ✓

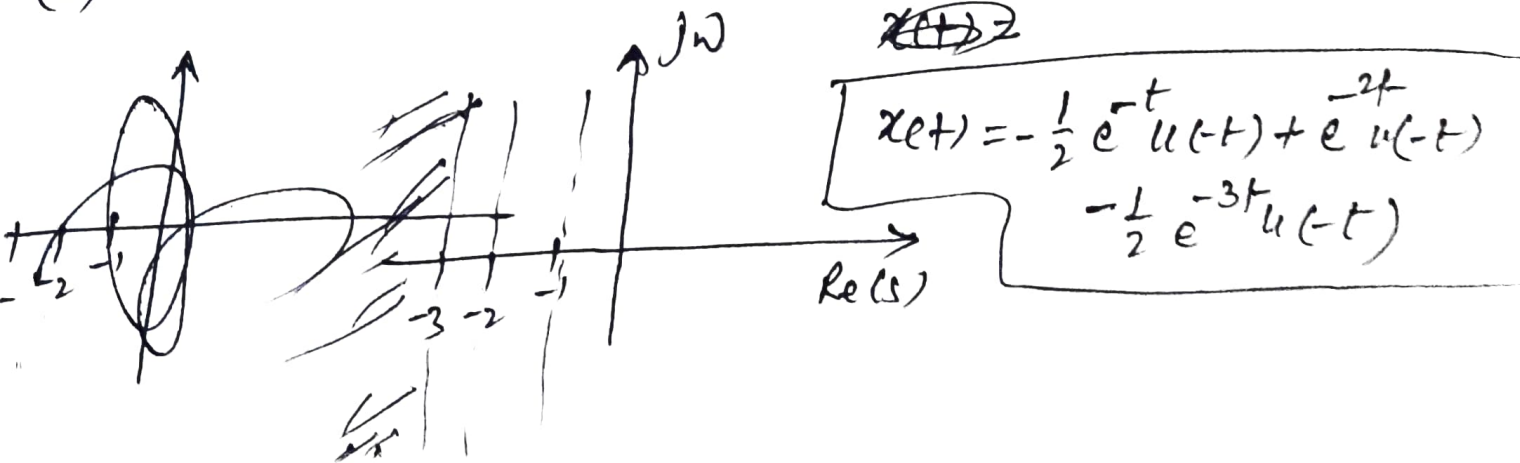
(iii) $\text{Re}(s) < -1$ ✗

(vii) $-2 < \text{Re}(s) < -1$

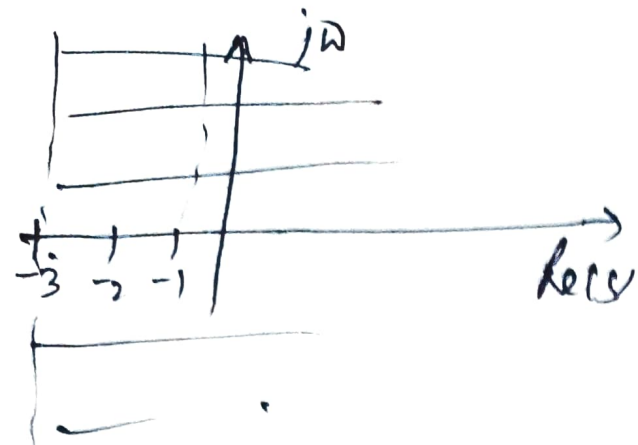
(iv) $\text{Re}(s) > -1$ ✓

(v) $-3 < \text{Re}(s) < -1$ ✗

(2') When $\text{Re}(s) < -3 \rightarrow$ ROC is towards left of most line



(ii) $\text{Re}(s) > -3 \rightarrow$ not possible. (13)

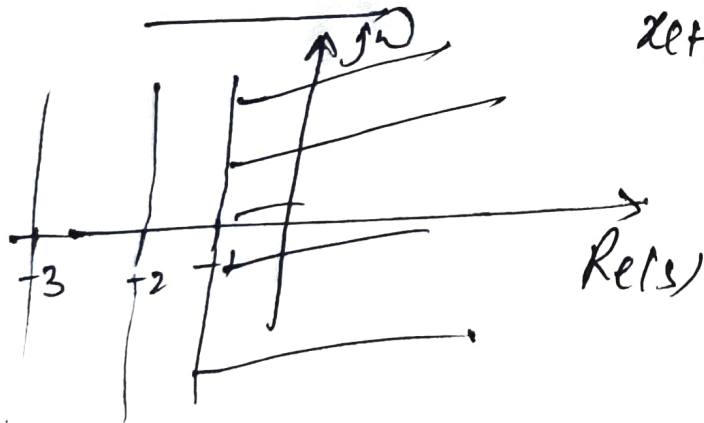


(iii) $\text{Re}(s) < -1 \rightarrow$ not possible



(iv)

$\text{Re}(s) > -1$



$$x(t) = \frac{1}{2} e^{-t} u(t) - e^{-2t} u(t) + \frac{1}{2} e^{-3t} u(t)$$

(v)

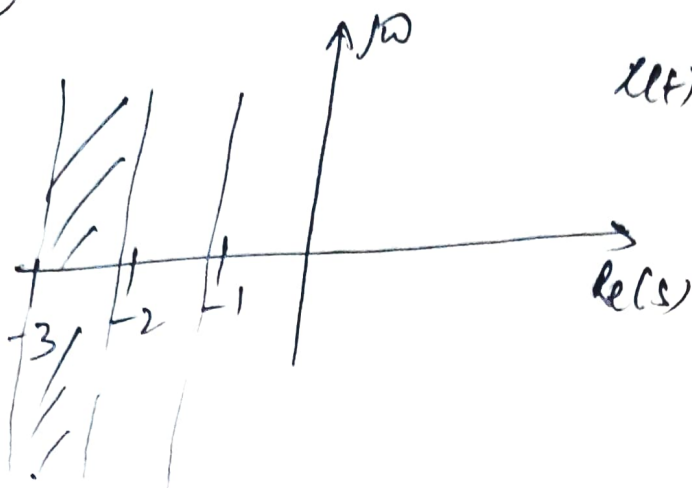
$-3 < \text{Re}(s) < -1 \rightarrow$ not possible



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(VI)

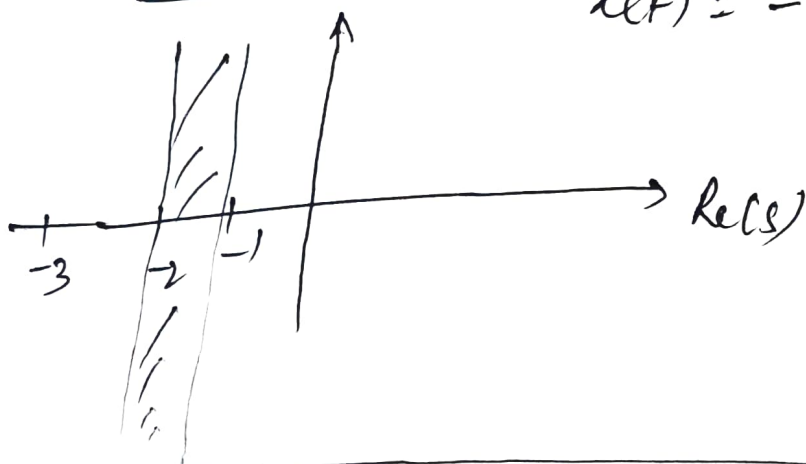
$$-3 < \operatorname{Re}(s) < -2$$



$$x(t) = -\frac{1}{2} e^{-t} u(-t) + e^{-2t} u(-t) + \frac{1}{2} e^{-3t} u(t)$$

(VII)

$$-2 < \operatorname{Re}(s) < -1$$



$$x(t) = -\frac{1}{2} e^{-t} u(-t) - e^{-2t} u(t) + \frac{1}{2} e^{-3t} u(t)$$

Ques

$$X(s) = \frac{2}{(s^2+4)(s+2)}, \quad \text{With ROC: } -$$

$$-2 < \operatorname{Re}(s) < 0$$

Soln:-

$$X(s) = \frac{2}{(s^2+4)(s+2)} = \frac{As+B}{s^2+4} + \frac{C}{(s+2)}$$

$$= \frac{(As+B)(s+2) + C(s^2+4)}{(s^2+4)(s+2)}$$

$$\text{So } (As+B)(s+2) + C(s^2+4) \equiv 2$$

→ Value of A, B, C

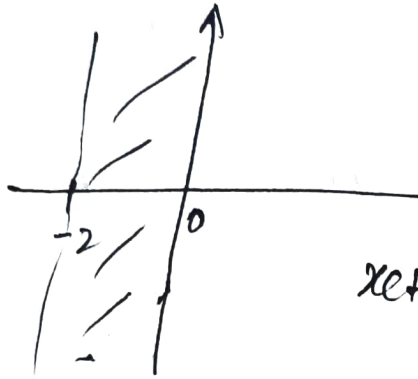
$$A = -\frac{1}{4}, \quad B = \frac{1}{2}, \quad C = \frac{1}{4}$$

So $X(s) = \frac{2}{(s^2+4)(s+2)} = \frac{-\frac{L}{4}s + \frac{L}{2}}{s^2+4} + \frac{\frac{L}{4}}{s+2}$

$= \frac{-\frac{L}{4}s}{s^2+4} + \frac{\frac{L}{2}(\frac{2}{2})}{s^2+4} + \frac{\frac{L}{4}}{(s+2)}$

ROT:

$-2 < \text{Re}(s) < 0$



Anti-causal

Causal

$x(t) = +\frac{L}{4} \cos 2t u(-t) - \frac{L}{4} \sin 2t u(-t) + \frac{L}{4} e^{-2t} u(t)$

Ques

find the Inverse Laplace transform of the following

$X(s) = \frac{s^2+2s+5}{(s+3)(s+5)^2}$

(a) $\text{Re}(s) < -5$

(b) $\text{Re}(s) > -3$

(c) $-5 < \text{Re}(s) < -3$

Soln:

$X(s) = \frac{s^2+2s+5}{(s+3)(s+5)^2} = \frac{A}{(s+3)} + \frac{B}{(s+5)} + \frac{C}{(s+5)^2}$

$= \frac{2}{(s+3)} - \frac{10}{(s+5)^2} - \frac{1}{(s+5)}$

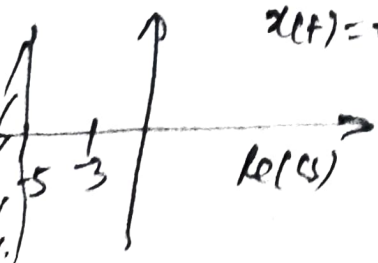
(b)

$\text{Re}(s) > -3$

$x(t) = 2e^{-3t} u(t) - 10te^{-5t} u(t) - e^{-5t} u(t)$

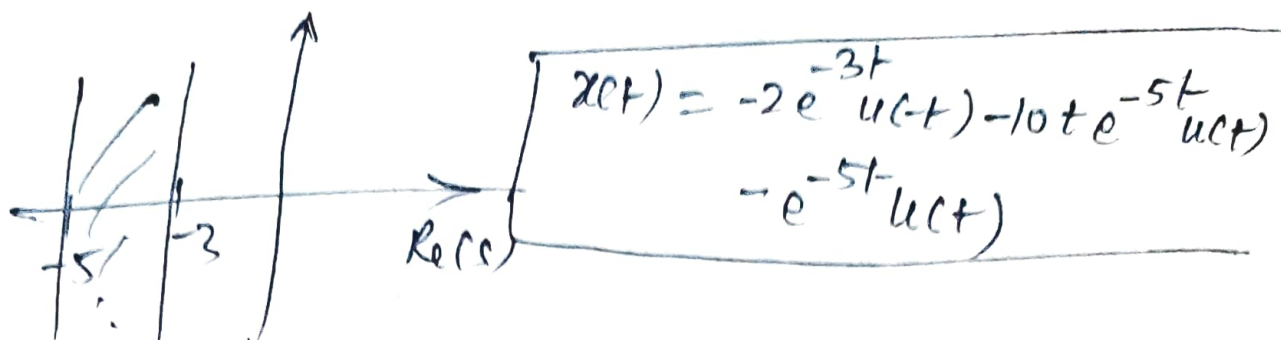
$x(t) = (2e^{-3t} - 10te^{-5t} - e^{-5t}) u(t)$

(a) $\text{Re}(s) < -5$



(C) Req: $-5 < \operatorname{Re}(s) < -3$

(16)



$$x(t) = -2e^{-3t}u(t) - 10te^{-5t}u(t) - e^{-5t}u(t)$$

Ques: Find all possible Inverse Laplace transform of

$$X(s) = \frac{1}{s(s+1)(s+2)(s+3)}$$

Soln:-

$$X(s) = \frac{1}{6} \left(\frac{1}{s} \right) - \frac{1}{2} \left(\frac{1}{s+1} \right) + \frac{1}{2} \left(\frac{1}{s+2} \right) - \frac{1}{6} \left(\frac{1}{s+3} \right)$$

(i) $\operatorname{Re}(s) > 0$

$$x(t) = \frac{1}{6}u(t) - \frac{1}{2}e^{-t}u(t) + \frac{1}{2}e^{-2t}u(t) - \frac{1}{6}e^{-3t}u(t)$$

(ii) $\operatorname{Re}(s) < -3$

$$x(t) = -\frac{1}{6}u(-t) + \frac{1}{2}e^{-t}u(-t) - \frac{1}{2}e^{-2t}u(-t) + \frac{1}{6}e^{-3t}u(-t)$$

(iii) $-1 < \operatorname{Re}(s) < 0$

$$x(t) = -\frac{1}{6}u(-t) - \frac{1}{2}e^{-t}u(t) + \frac{1}{2}e^{-2t}u(t) - \frac{1}{6}e^{-3t}u(t)$$

(iv) $-2 < \operatorname{Re}(s) < -1$

$$x(t) = -\frac{1}{6}u(-t) + \frac{1}{2}e^{-t}u(-t) + \frac{1}{2}e^{-2t}u(t) - \frac{1}{6}e^{-3t}u(t)$$

(v) $-3 < \operatorname{Re}(s) < -2$

$$x(t) = -\frac{1}{6}u(-t) + \frac{1}{2}e^{-t}u(-t) - \frac{1}{2}e^{-2t}u(-t) - \frac{1}{6}e^{-3t}u(t)$$